3.2 - Mean Value Theorem

Goals: State and Apply Rolle's Theorem and the Mean Value Theorem
Before Class Videos: 3.2.1

Note: Friday's class will continue with the prior 3.1 worksheet before moving to this one.

Theorem (Rolle's Theorem):
If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ so $f'(c) = 0$.

Proof:

Theorem (Mean Value Theorem):
Suppose $f(x)$ is continuous on $[a, b]$ anddifferentiable on (a, b).
Then there exists c in (a, b) so

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

Example 1: $y = \sqrt{x-2}$ on $[2, 6]$.

Example 2: Suppose $f(1) = 10$ and $f'(x) \geq 2$ on $1 \leq x \leq 4$. What is a minimum possible value of $f(4)$?

Example 3: $f(x) = x^3 + 3x + 1$. Show exactly one root.

Theorem:
Suppose $f'(x) = 0$ on $[a, b]$. Then $f(x)$ is constant.

Corollary: Suppose $f'(x) = g'(x)$ on (a, b). Then $f(x) = g(x) + C$ for some constant.

Example 4: If $f'(x) = x^2$, what is $f(x)$?

Suggested after class problems from APEX: Section 3.2: 1,5,11